

Al-Aqsa University

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جامعة الأقصى
كلية الحاسبات وتكنولوجيا المعلومات
قسم الهندسة والشبكات

Operating Systems for Intelligent Systems (ENGS2306)

Lecture 4: Linux Users and Groups Management

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User IDs and Their Indications

- **What are User IDs (UIDs)?:**
 - Unique numeric identifiers for users in Linux.
 - Stored in /etc/passwd (e.g., student:x:1000:1000).
 - **Group IDs (GIDs):**
 - Numeric identifiers for groups, stored in /etc/group.
 - Users belong to a primary group and optional secondary groups.
 - **Indications:**
 - **UID 0:** Root (superuser) with unrestricted access.
 - **UID 1–999:** System users (e.g., www-data for web services).
 - **UID $\geq$ 1000:** Regular users (e.g., student).
 - **Viewing IDs:**
 - `id`: Shows UID, GID, and groups (e.g., uid=1000(student) gid=1000(users)).
 - `cat /etc/passwd`: Lists users with UIDs.
 - **Relevance:**
 - UIDs/GIDs determine file ownership and permissions in intelligent systems.
 - Example: IoT device restricts config files to root (UID 0).
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Linux User Types

- **User Categories:**
 - **Root User:**
 - UID 0, full system access.
 - Used for critical tasks (e.g., `sudo apt update`).
 - **Regular Users:**
 - UID $\geq$ 1000, limited permissions for security.
 - Example: student for user-specific tasks.
 - **System Users:**
 - UID 1–999, created for services (e.g., www-data for web servers).
 - No login access, restricted roles.

- **Why It Matters:**
 - User types ensure secure access control in multi-user systems.
 - Root manages IoT hardware; regular users handle app data.
 - **Example:**
 - `cat /etc/passwd | grep www-data`: Shows system user for web server.
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Creating and Deleting Users

- **Creating Users:**
 - `sudo useradd -m username`: Creates user with home directory.
 - `-m`: Creates `/home/username`.
 - `sudo passwd username`: Sets user password.
 - Example: `sudo useradd -m guest`; `sudo passwd guest`.
 - **Deleting Users:**
 - `sudo userdel username`: Removes user, leaves home directory.
 - `sudo userdel -r username`: Removes user and home directory.
 - Example: `sudo userdel -r guest`.
 - **Verification:**
 - `cat /etc/passwd | grep username`: Confirms user exists or deleted.
 - **Use Case:**
 - Create temporary users for IoT testing, delete after use.
 - **Tip:** Use `sudo` for user management commands.
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Changing User Passwords and Usernames

- **Changing Passwords:**
 - `sudo passwd username`: Updates user password.
 - Example: `sudo passwd student` prompts for new password.
 - For current user: `passwd`.
- **Changing Usernames:**
 - `sudo usermod -l newname oldname`: Renames user.

- `-l`: Specifies new login name.
 - Update home directory: `sudo usermod -d /home/newname -m newname`.
 - Example: `sudo usermod -l newstudent student; sudo usermod -d /home/newstudent -m newstudent`.
 - **Verification:**
 - `id newname`: Confirms new username and UID.
 - `cat /etc/passwd`: Checks updated entry.
 - **Use Case:**
 - Update passwords for security in Android-based systems.
 - Rename users during system reconfiguration.
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Managing Groups and Adding Users to Groups

- **Creating Groups:**
 - `sudo groupadd groupname`: Creates a new group.
 - Example: `sudo groupadd developers`.
 - **Deleting Groups:**
 - `sudo groupdel groupname`: Removes group (if no users assigned).
 - Example: `sudo groupdel developers`.
 - **Adding Users to Groups:**
 - `sudo usermod -aG groupname username`: Adds user to group.
 - `-aG`: Appends to secondary groups.
 - Example: `sudo usermod -aG developers student`.
 - **Verification:**
 - `groups username`: Lists user's groups.
 - `cat /etc/group | grep groupname`: Confirms group and members.
 - **Use Case:**
 - Assign IoT developers to a group for shared access to config files.
 - **Tip:** Use `-a` to avoid overwriting existing groups.
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Practical Users and Groups Scenario

- **Task:** Manage users and groups in Ubuntu.
 1. Open terminal (Ctrl+Alt+T).
 2. Check current user: id.
 3. Create user: sudo useradd -m testuser; sudo passwd testuser.
 4. Verify: cat /etc/passwd | grep testuser.
 5. Change password: sudo passwd testuser.
 6. Rename user: sudo usermod -l newuser testuser; sudo usermod -d /home/newuser -m newuser.
 7. Create group: sudo groupadd testgroup.
 8. Add user to group: sudo usermod -aG testgroup newuser.
 9. Verify: groups newuser; cat /etc/group | grep testgroup.
 10. Delete user and group: sudo userdel -r newuser; sudo groupdel testgroup.
 - **Goal:** Practice user and group management for system administration.
 - **Record:** Save outputs for Assignment 1.
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Idea Behind the Assignment

- **Learning Objectives:**
 - **Understand UIDs/GIDs:** Analyze user and group IDs to grasp their role in ownership and permissions.
 - **Master User Management:** Create, delete, and modify users to simulate secure access control in intelligent systems.
 - **Apply Group Management:** Manage groups to organize users and streamline permissions, critical for multi-user environments.
 - **Enhance Security:** Practice password updates and user renaming to secure IoT and Android systems.
- **Why It Matters:**
 - User/group management ensures secure access in intelligent systems (e.g., IoT device admins, Android app users).
 - Prepares for Week 13's Raspberry Pi OS configuration.

- **Practical Insight:** Simulates administering a multi-user system, balancing security and access for collaborative projects.
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Week 3 Assignment

- **Assignment 1:** Manage Users, Groups, and Permissions

- **Tasks:**

1. **Explore Users and Groups:**

- Run `id` to display current user's UID, GID, and groups.
- Run `cat /etc/passwd | grep -E '^(root|users|[^:]*:x:[0-9]{4,})'` to list root and regular users (UID $\geq$ 1000).
- Run `cat /etc/group | grep users` to view group details.
- Identify one system user (e.g., `www-data`) and explain its purpose (e.g., web server).

2. **Manage Users:**

- Create: `sudo useradd -m 1001 testuser1; sudo passwd testuser1.`
- Create another: `sudo useradd -m testuser2; sudo passwd testuser2.`
- Change password: `sudo passwd testuser1.`
- Rename: `sudo usermod -l newuser2 testuser2; sudo usermod -ld /home/newuser2 -m newuser2.`
- Verify: `cat /etc/passwd | grep newuser2.`

3. **Manage Groups:**

- Create: `sudo groupadd projectgroup.`
- Add users: `sudo usermod -aG projectgroup testuser1; sudo usermod -aG projectgroup newuser2.`
- Verify: `groups testuser1; cat /etc/group | grep projectgroup.`

4. **Apply Permissions:**

- Create: `mkdir ~/project_dir; touch ~/project_dir/doc.txt.`
- Set permissions: `chmod 660 ~/project_dir/doc.txt (owner: rw, group: rw).`
- Set ownership: `chown testuser1:projectgroup ~/project_dir/doc.txt.`
- Verify: `ls -l ~/project_dir.`
- **Analyze:**

- Attempt to access doc.txt as newuser2 (if possible, e.g., via `sudo su newuser2`) and document result.
- Explain why 660 allows group access.

5. Cleanup:

- Delete: `sudo userdel -r testuser1`; `sudo userdel -r newuser2`; `sudo groupdel projectgroup`.
- Verify: `cat /etc/passwd`; `cat /etc/group`.

6. Submit:

- Compile outputs (screenshots or text, e.g., `ls -l > output.txt`).
- Write a brief report (1–2 paragraphs) explaining one user operation (e.g., renaming) and one group benefit (e.g., shared access).
- Submit as PDF or text via Moodle.

- **Tips:**

1. Use `man useradd`, `man usermod`, `man groupadd` for help.
2. Save outputs: `id > output.txt`.
3. Take screenshots with `PrtSc` key.

- **Deadline:** End of Week 3.
- **Weight:** 12% of practical grade.

Next Steps

- **Complete Tasks:**

- Practice `id`, `useradd`, `userdel`, `passwd`, `usermod`, `groupadd`, `groupdel` in Ubuntu.
- Experiment with permissions (e.g., `chmod 660 file`, `chown user:group file`).
- Save snapshots in VMware to revert changes.

- **Review:**

- Rewatch Lecture 4 video on YouTube/Moodle.
- Use man pages or Ubuntu forums for command details.

- **Prepare for Week 4:** Process Management in Linux (`ps`, `top`, `htop`).

- **Support:** Contact Eng. Mohammed E. Alhams via Moodle or the course coordinator.
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Additional Resources

- **Ubuntu Documentation:** <https://help.ubuntu.com>
- **Linux Users Guide:** https://tldp.org/LDP/intro-linux/html/sect_03_02.html
- **User Management Tutorial:** <https://www.digitalocean.com/community/tutorials/how-to-manage-users-in-ubuntu>
- **Group Management Guide:** <https://www.tecmint.com/manage-users-and-groups-in-linux/>
- **Moodle:** Access lecture videos, ISOs, and assignment submission.
- **Tip:** Download resources during stable internet hours in Gaza to avoid connectivity issues.